

Rahul Shah

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EDUCATION

University of California, Berkeley

GPA: 4.0

M.S. Electrical Engineering & Computer Science. Thesis: *Optimizing ML Systems*

May 2025

B.A. Computer Science; B.A. Mathematics; B.A. Statistics

August 2020 - May 2024

Honors: Dean's List, Honors to Date, CS Honor Society (UPE) Officer, EECS Honors, Higher Honors in Mathematics

Activities & Societies: CSUA (CS Club Executive Officer), CS Mentors (CS 61A/B/70/88 Tutor), IEEE, ICPC.

Relevant Coursework: Data Structures & Algorithms[†], Natural Language Processing[†], Programming Languages & Compilers[†], Machine Learning, Neural Nets[†], Deep Reinforcement Learning[†], Data Mining, Parallel Systems[†], Computer Vision[†], Forecasting[†], Human-Computer Interaction[†], Computer Graphics[†], Probability Theory[†], Optimization Methods[†]. [†]= Graduate

EXPERIENCE

• Snap Inc.

San Francisco, CA

Machine Learning Engineer Intern

Jan 2025 - May 2025

- Quantized FLUX and ControlNet models with SVD/OpenFLUX to optimize memory and preserve image quality.
- Benchmarked and improved IP-Adapter pipelines with VAE compression and dynamic aspect ratio tuning.

• Schonfeld Strategic Advisors (Quantitative Hedge Fund)

New York, NY

Applied Machine Learning Engineer (Alpha **Quantitative Researcher**) Intern

Apr 2023 - Aug 2023

- Optimized >100 GB of alternative data from AWS S3 & EC2. Exploratory Data Analysis using **Databricks**' PySpark, for a statistically backtested scalable signal (currently in production) with Sharpe Ratio 1.8. Under Systematic Trading team (QIS).
- Wrote an **Online** Algorithm to ingest live data & improve latency by 53% via Statistical Estimation for a HFT Stat Arb strategy. Measured effects of out-of-sample distributional shift via **divergence** measures, for a 12x improvement.

• Arrowstreet Capital LP | Research Core

Boston, MA

Software Development Engineer Intern @ Arrowstreet's Team

Jun 2022 - Aug 2022

- Fine-tuned a local **FinBERT** model from **Hugging Face** on over 100,000 vendor filings of various languages & correlated sentiments with word embeddings using **Cosine Similarity**, Jaccard Index, and Minkowski Distances.
- Wrote distributed code with **Numba** and **Dask** to discover hidden signals for equity alpha research. Made robust improvements to the firm-wide single systematic process with equities toolings via **Bayesian Regressions** saving \$15 million annually.

• Electrical Engineering & Computer Sciences | University of California, Berkeley

Berkeley, CA

Researcher; Teaching: Instructor, Head Teaching Assistant, Graduate Student Instructor.

Jan 2021 - Present

- Summer 2024 Instructor - CS 160 (Human-Computer Interaction):** Lectured Contextual Inquiry, creating user flows, wireframes, and building user interface mockups/prototypes. Advanced Javascript coding and frameworks.
- Lead TA - CS C100/200 (Data Science/ML), STAT 165/265 (Forecasting), CS 164 (Compilers), CS 267 (Parallel Systems):** Managed admin logistics for ~1000 students, automated assignment extensions via Gradescope API.
- Head Teaching Assistant - EECS 122 (Communication Networking/Internet):** Lead all administrative course logistics, created all existing course content, held office hours and led section on-campus.

- Berkeley Research - Artificial Intelligence:** Deep Learning & NLP: Trained an **LSTM** to predict student grades from attendance and analyzed **Time Series** to determine earnestness with new SOTA **metrics** (Paper accepted in ACM's 2023 conference: <https://arxiv.org/abs/2311.10746>). Also applied Deep RL techniques to 'solve' Minecraft.

PROJECTS

- Live Streaming NeRF Nerfstudio Plugin:** Live re-rendering of shaders and textures in a game environment in real-time with **minimal latency**, with efficient C++ code, as well as adaptive noise reduction with anisotropic filters. Used a novel networking protocol to parse and transmit the live data from PyTorch, without loss or unmatched data pairs.
- GPU-accelerated Factorization Machines:** Boosted recommender systems training speed by **21.4%** using a novel randomized algorithm that capitalized on data **sparsity** to parallelize Stochastic Gradient Descent. Achieved via custom CUDA structures for efficient **low-rank** polynomial kernel approximation. Profiled bottlenecks with VTune's Roofline.

AWARDS

- Donald M. Rizzo Memorial Scholarship Recipient:** Awarded from international NPO: Lions Club International
- Winner of the Congressional App Challenge:** Invited to Capitol Hill to demo mobile app to House of Representative members at #HouseOfCode, the most prestigious student STEM expo, at the U.S. Capitol Building
- Capture The Flag Security Competitions:** Ranked **top 50** in all of {PicoCTF, HSCTF, BACTF, CSAW Red}

SKILLS

- Languages:** Python, Java, Go, C++, JavaScript/Node.js, Rust, Scala, C, TypeScript, Objective-C, C#, Shell (bash, zsh), OCaml, R.
- Technologies:** AWS, Databricks, Heroku, GCP; Docker/Kubernetes; Parquet, MySQL/PostgreSQL, MongoDB; React, Vue.
- AI/ML:** PyTorch, Keras, Tensorflow; LASSO, ANOVA, Kalman Filters; CNN, R-CNN, GNN, Transformer, BERT, SSM, RAG.